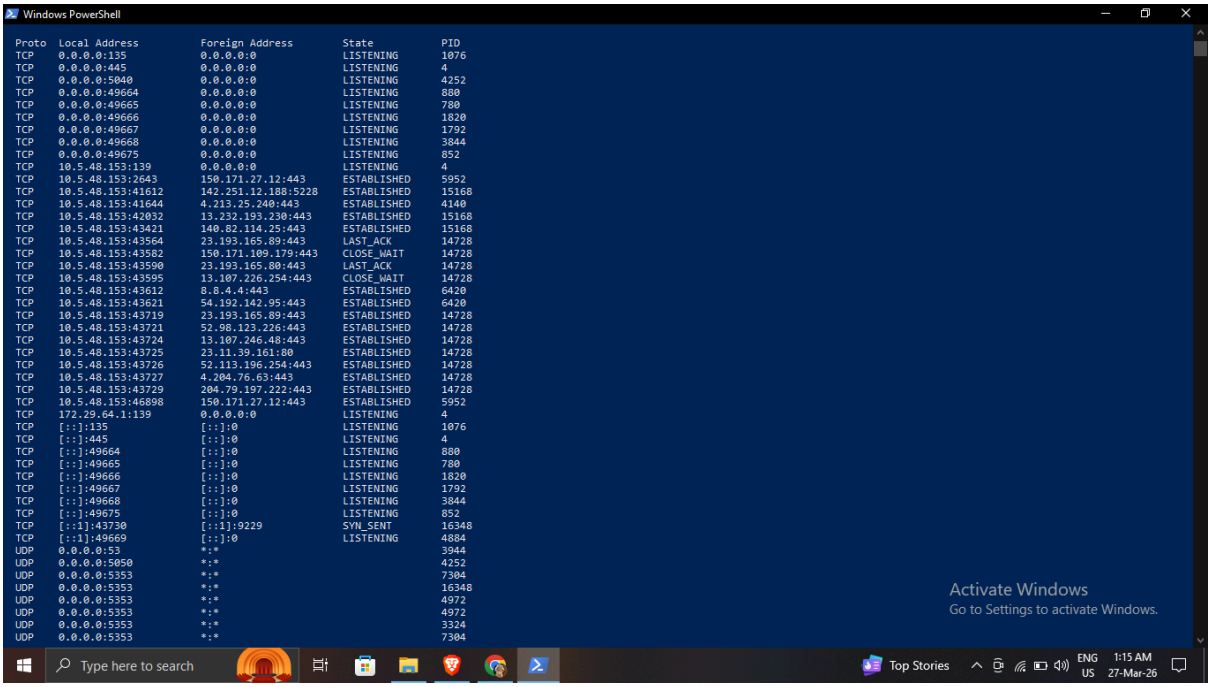
Capstone Project (Full IR Cycle)

A. Forensic Evidence Collection (Volatile Data)

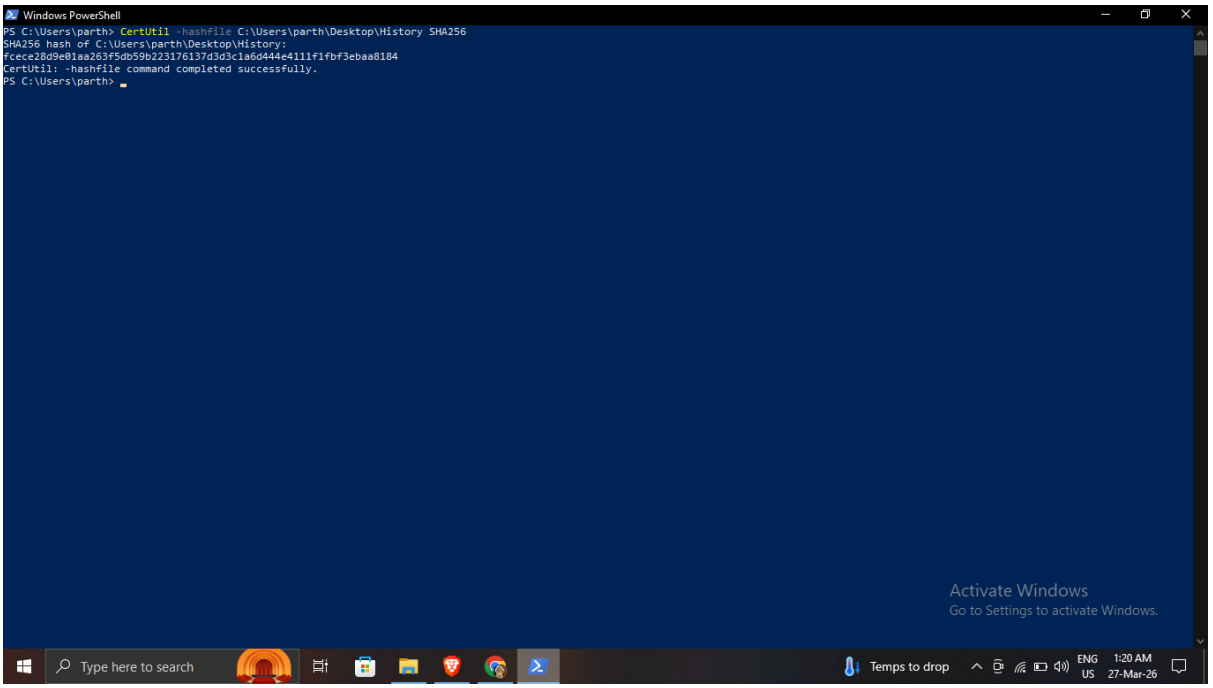
To identify the source of unauthorized activity, volatile network data was collected from the host machine.



- **Action:** Used the `netstat -ano` command to list all active TCP/UDP connections and their associated Process IDs (PIDs).
- **Analysis:** This allowed for the identification of remote IP addresses communicating with the host.

B. Evidence Preservation (Data Integrity)

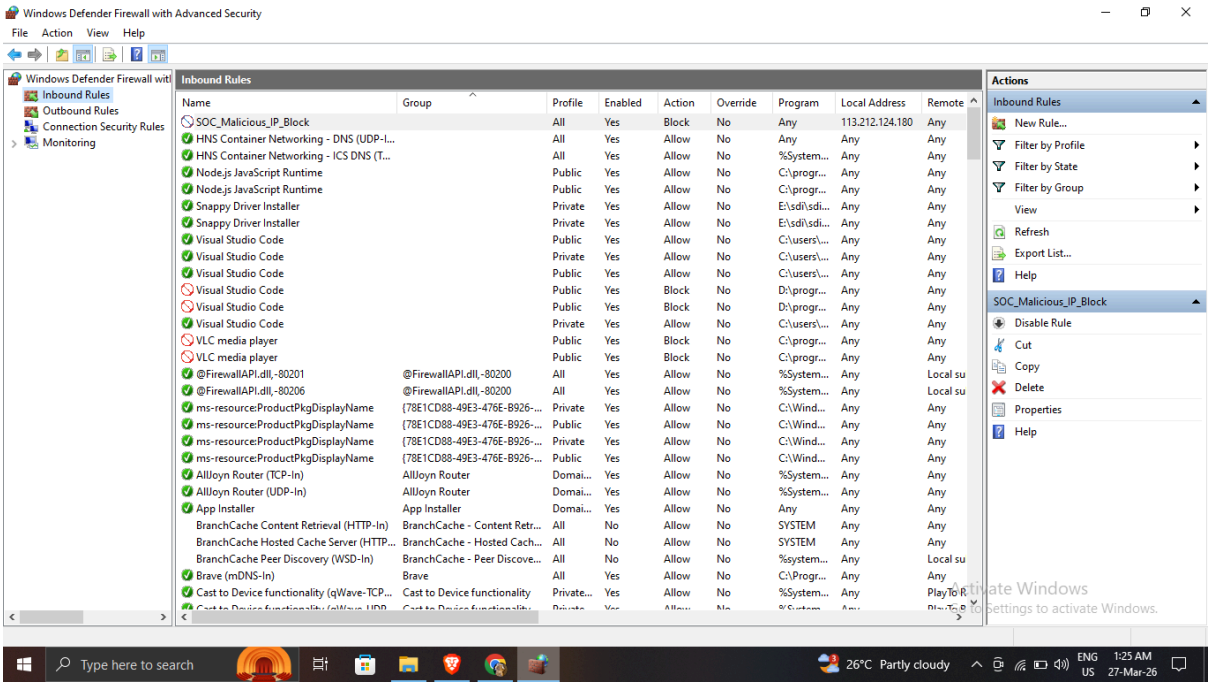
To maintain the chain of custody, a cryptographic hash was generated for the identified evidence files.



- **Action:** Generated a **SHA256** hash using the CertUtil utility.
- **Result:** The unique hash fcece28... ensures that the evidence remains untampered throughout the investigation lifecycle.

C. Containment & Response (Host-Based Firewall)

Once the malicious source IP was validated via Threat Intelligence, immediate containment was performed to stop further data exfiltration.



- **Action:** Configured a new Inbound Firewall Rule named `SOC_Malicious_IP_Block`.
- **Mechanism:** The rule is set to **"Block"** all traffic from the remote IP `113.212.124.180`.
- **Outcome:** The threat was successfully isolated from the system without interrupting legitimate network services.

4. Final Incident Report (SANS Template Summary)

Executive Summary: A simulated security incident was detected and contained on the analyst's workstation. The incident involved an unauthorized connection attempt from a known malicious IP address associated with phishing activities.

Timeline (March 27, 2026):

- **01:15 AM:** Identification of suspicious network socket via `netstat`.
- **01:20 AM:** Forensic hashing completed for evidence preservation.
- **01:25 AM:** Containment successfully implemented via Windows Firewall.

Technical Analysis: The investigation successfully mapped network connections to system processes. Threat Intelligence lookup on VirusTotal confirmed the external IP had a high malicious detection rate (8/95).

Remediation & Lessons Learned: The immediate threat was mitigated by blacklisting the source IP. **Recommendation:** Implement automated Endpoint Detection and Response (EDR) to reduce the Mean Time to Respond (MTTR) in future incidents.